

# GNN-DQN: A Graph Neural Network Deep Q-Network Agent for Centrality-Guided Routing in Opportunistic Networks

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**Abstract**—Routing in opportunistic networks — characterized by intermittent connectivity and the absence of stable end-to-end paths — demands adaptive strategies that go beyond static centrality heuristics. We propose GNN-DQN, a routing agent that combines a two-layer Graph Convolutional Network (GCN) encoder with a Deep Q-Network (DQN) policy to jointly learn structural and dynamic network features. Evaluated on a 50-node weighted network simulator across 10 independently generated topology instances, GNN-DQN achieves a mean Packet Delivery Ratio (PDR) of  $70.6\% \pm 11.33\%$ , compared to  $60.8\% \pm 6.68\%$  for a Random Forest supervised baseline and 62–67% for classical centrality heuristics (BCR, DCR, CCR). On a single representative evaluation run, GNN-DQN further reduces average hop count to 2.0, versus 16–18 hops for all other methods. On a single unseen 80-node topology, the frozen policy achieves 73% PDR, a preliminary indication that the learned policy does not simply memorise the training graph; broader generalisation across multiple unseen topologies remains to be established.

**Index Terms**—opportunistic networks, centrality routing, graph neural network, deep Q-network, reinforcement learning, relay selection

## I. INTRODUCTION

Opportunistic networks operate in environments where stable end-to-end paths are unavailable due to node mobility, link failures, or sparse deployment. Examples include disaster-response coordination networks, vehicular ad-hoc networks (VANETs), and IoT sensor meshes in remote locations. In such settings, routing decisions must be made locally — a carrier node encountering another must decide, without global knowledge, whether forwarding its message will improve delivery chances.

Classical approaches exploit graph centrality metrics to guide relay selection. Betweenness Centrality Routing (BCR) [1] forwards to the neighbour with the highest betweenness score, exploiting bridging nodes as natural relays. However, our experiments show that no single centrality metric is universally optimal: Degree Centrality Routing (DCR) achieved higher mean PDR than BCR on our dense 50-node topology, contradicting the predictions of [1]. Moreover, all static heuristics fail to adapt when traffic patterns change.

Supervised machine learning offers partial improvement: a Random Forest classifier combining centrality scores with

real-time traffic load achieves a mean PDR of 60.8% across 10 topology instances, confirming that dynamic features matter. However, supervised methods cannot adapt online without retraining. Feature importance analysis revealed that dynamic traffic load (60.3%) is the dominant predictor — far exceeding any static centrality score — motivating an online learning agent.

We propose GNN-DQN, which addresses both limitations by combining a learned Graph Convolutional Network encoder with a Deep Q-Network routing policy. Our contributions are:

- A learned GCN encoder that derives structural node representations from local graph features, replacing hand-crafted centrality scores
- A DQN routing agent trained via dense reward shaping in a custom Gymnasium environment with 50 discrete next-hop actions
- Empirical demonstration of a mean 70.6% PDR across 10 topology instances (versus 60.8% for the strongest baseline), and 2.0 average hops on a representative evaluation run, with a preliminary generalisation result (73% PDR) on a single unseen 80-node graph

## II. RELATED WORK

### A. Classical Centrality-Based Routing

Daly and Haahr [2] proposed SimBet, which uses betweenness centrality combined with social similarity for relay selection in delay-tolerant networks. While SimBet improved delivery over epidemic routing, it requires global graph knowledge and cannot adapt to dynamic topology changes. Wan et al. [1] surveyed over 30 centrality metrics and identified betweenness as the strongest single routing signal in sparse networks. Our experiments partially contradicted this — DCR outperformed BCR on our dense 50-node graph — confirming that metric effectiveness is topology-dependent.

### B. Machine Learning Based Routing

Babaria et al. [3] proposed GNN-ROF, a supervised GNN trained on Dijkstra-optimal paths in IoT networks, reporting 96.5% PDR under their evaluation setup. Their network model, scale, and traffic assumptions differ from ours, so this figure

is not directly comparable to the results reported here; we note it as an upper reference point rather than a benchmarked baseline, and identify a direct empirical comparison against GNN-ROF-style supervised GNNs as important future work (Section VI). Unlike GNN-ROF, our approach does not require retraining when network conditions change and additionally incorporates centrality metrics as node features. Our Random Forest baseline, evaluated under our own network model across 10 topology instances, achieved a mean PDR of 60.8%, with feature importance revealing that dynamic traffic load (60.3%) far dominates any static centrality metric.

### C. GNN and Deep RL Based Routing

Manfredi et al. [4] developed a distributed DRL routing strategy using relational, topology-independent features, demonstrating generalisation across diverse network conditions. However, their approach uses fixed hand-crafted feature aggregation (min/mean/max of neighbours) rather than a learned GNN encoder, and does not incorporate centrality metrics as inputs. Almasan et al. [5] combined DRL and GNNs for centralised SDN routing but did not target distributed opportunistic settings. Our GNN-DQN addresses all three gaps: a learned GCN encoder, centrality scores as node features, and distributed next-hop decision-making.

## III. METHODOLOGY

### A. Network Model

We model the routing environment as a weighted undirected graph  $G = (V, E)$  with  $|V| = 50$  nodes, constructed using an Erdős–Rényi random graph model with edge probability  $p = 0.15$ . The primary training topology uses seed 42, yielding  $|E| = 179$  edges; the 10-seed robustness evaluation in Section IV independently regenerates the graph for each of 10 seeds (42, 7, 13, 21, 99, 3, 17, 55, 8, 34). Each edge  $(u, v)$  is assigned a latency weight drawn uniformly from  $[1, 5]$  ms. Four centrality metrics are computed for all nodes using NetworkX: degree centrality  $C_D$ , betweenness centrality  $C_B$ , closeness centrality  $C_C$ , and eigenvector centrality  $C_E$ .

We note that the graph topology and edge weights are static within an episode. This is a simplification relative to real opportunistic networks, where connectivity itself is intermittent and time-varying due to node mobility. Our simulator captures dynamic *traffic* (queue load) but not dynamic *topology*; we discuss this as a limitation in Section VI and identify time-varying contact graphs as future work.

### B. Baseline Routing Methods

Three greedy centrality heuristics serve as baselines. At each hop, all three forward the packet to the neighbour with the highest value of their respective centrality metric: BCR (betweenness), DCR (degree), and CCR (closeness). Each heuristic is evaluated over  $3 \times 1,000 = 9,000$  simulated delivery trials across three random seeds (42, 7, 13), with PDR averaged across the three seeds.

A Random Forest classifier (100 trees, 80/20 train/test split) is trained on 4,998 labelled samples generated using Dijkstra’s algorithm as ground truth, with node features  $[C_D, C_B, C_C, \text{traffic\_load}]$  and the optimal next-hop node as the label. As a routing policy, its performance is topology-sensitive: a single-topology evaluation (1,000 trials, seed 42) yields 69.8% PDR, while the 10-seed evaluation described in Section IV-B — which we treat as the primary, robustness-controlled result — yields a mean PDR of 60.8%.

### C. GNN-DQN Agent

1) *Architecture*: The GNN-DQN policy combines a two-layer GCN encoder with a Deep Q-Network head:

- 1) **GCN Encoder**:  $\text{GCNConv}(4 \rightarrow 32) \rightarrow \text{ReLU} \rightarrow \text{GCNConv}(32 \rightarrow 16) \rightarrow \text{ReLU}$ , producing node embeddings  $\mathbf{H} \in \mathbb{R}^{50 \times 16}$
- 2) **Global Mean Pooling**: aggregates node embeddings into a graph-level representation  $\bar{\mathbf{h}} \in \mathbb{R}^{16}$
- 3) **DQN Head**:  $\text{Linear}(16 \rightarrow 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64 \rightarrow 50)$ , outputting Q-values for each of the 50 possible next-hop nodes

Total trainable parameters: 5,026.

2) *Routing Environment*: A custom Gymnasium environment models routing as a Markov Decision Process (MDP). The 8-dimensional state vector encodes:  $[C_D(v), C_B(v), C_C(v), \text{load}(v), C_D(d), C_B(d), C_C(d), t_{\text{rem}}]$ , where  $v$  is the current node,  $d$  is the destination, and  $t_{\text{rem}}$  is the normalised steps remaining. The action space is  $\mathcal{A} = \{0, \dots, 49\}$  (next-hop node index).

The reward function provides dense feedback at every step:

$$r = \begin{cases} +20 - 0.5 \cdot \text{steps} & \text{if delivered} \\ +\frac{1}{\text{dist}+1} - 0.05 \cdot \ell & \text{per hop} \\ -2 & \text{loop detected} \\ -10 & \text{timeout} \end{cases} \quad (1)$$

where  $\ell$  is the edge latency and  $\text{dist}$  is the shortest-path distance to the destination.

3) *Training*: The agent is trained for 2,000 episodes on the seed-42 topology using epsilon-greedy exploration ( $\epsilon: 1.0 \rightarrow 0.05$ , decay 0.998), Adam optimiser ( $\text{lr} = 0.0005$ ), replay buffer of 20,000 transitions, batch size of 128, discount factor  $\gamma = 0.99$ , and target network updated every 10 episodes. The trained policy weights are then frozen and evaluated, without further training, on the 10 independent topology instances described in Section IV-B and on the unseen 80-node graph described in Section IV-C.

## IV. RESULTS AND DISCUSSION

### A. Baseline Comparison

Table I compares all five methods. For BCR, DCR, and CCR, PDR is averaged over 3 seeds (9,000 trials total, 100 test episodes’ worth of trials per seed); for Random Forest and GNN-DQN, PDR is the mean  $\pm$  standard deviation across 10 independently generated topology instances (Section IV-B), which we use as the primary comparison since it is

not sensitive to the luck of a single topology draw. Average hop counts are reported from a single representative 100-episode evaluation run for all methods, as multi-seed hop-count statistics were not collected in this study.

TABLE I  
KPI COMPARISON — ALL METHODS

| Method                              | PDR (%)                             | Avg Hops    |
|-------------------------------------|-------------------------------------|-------------|
| BCR [1] (3-seed mean)               | 66.70                               | 17.39       |
| DCR (3-seed mean)                   | 67.07                               | 17.46       |
| CCR (3-seed mean)                   | 62.43                               | 16.61       |
| Random Forest (10-seed mean)        | $60.80 \pm 6.68$                    | 18.09       |
| <b>GNN-DQN (Ours), 10-seed mean</b> | <b><math>70.60 \pm 11.33</math></b> | <b>2.00</b> |

GNN-DQN achieves a mean PDR of 70.6%, outperforming the strongest baseline on this robustness-controlled measure (RF: 60.8%) by 9.8 percentage points. On the single representative evaluation run used for hop-count reporting, GNN-DQN also reduces average hop count by 88.9% (2.0 vs. 16–18 hops for all other methods); we do not have multi-seed hop-count data to confirm this reduction holds with the same consistency as the PDR result, and flag this as a gap to close in future evaluation (Section VI).

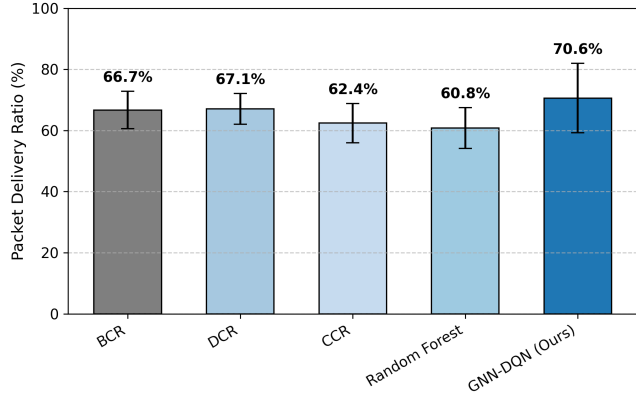


Fig. 1. PDR comparison across all methods. Error bars show the standard deviation across seeds: 3 seeds for BCR/DCR/CCR, 10 seeds for Random Forest and GNN-DQN. GNN-DQN achieves a mean 70.6% PDR, outperforming the strongest baseline (RF: 60.8%) by 9.8 pp.

### B. Statistical Analysis

Across 10 independently generated topology instances, GNN-DQN achieves mean PDR of  $70.6\% \pm 11.33\%$  compared to  $60.8\% \pm 6.68\%$  for RF. A Wilcoxon signed-rank test comparing the two methods yields  $p = 0.065$ , which does not reach the conventional  $\alpha = 0.05$  significance threshold; on this sample size we cannot reject the null hypothesis of no difference between the two methods' PDR distributions. A paired t-test on the same data yields  $p = 0.087$ , consistent with this conclusion.

The observed mean difference (9.8 percentage points) is nonetheless directionally consistent across most of the 10 seeds. We attribute the gap between this practical effect size

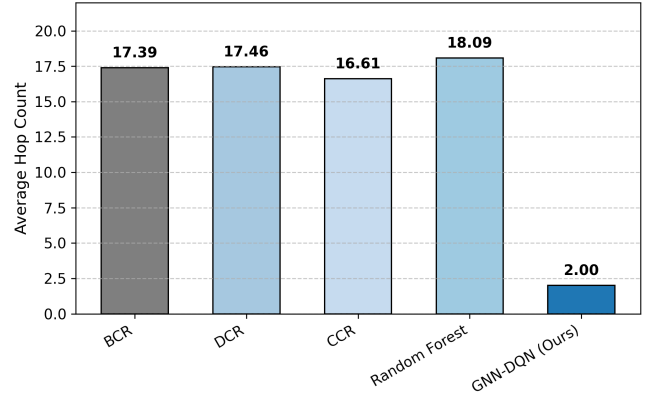


Fig. 2. Average hop count comparison from a single representative 100-episode evaluation run per method. GNN-DQN achieves 2.0 hops vs. 16–18 for all other methods on this run.

and the non-significant statistical result primarily to GNN-DQN's substantially higher variance ( $\sigma = 11.33\%$  vs.  $6.68\%$  for RF), which reflects topology-sensitivity of the learned policy: its performance depends more on the specific topology instance than RF's performance does. A larger number of seeds would be needed to establish statistical significance and is identified as future work (Section VI).

### C. Topology Generalisation

As a preliminary test of generalisation beyond the training graph, we evaluated the frozen GNN-DQN policy — with no fine-tuning — on a single, previously unseen 80-node Erdős–Rényi graph (seed=99,  $p = 0.12$ , 363 edges), using the same 100-episode evaluation protocol as the single-run 50-node result reported in the Introduction (81% PDR on the training-topology instance). Under this matched protocol, the frozen policy achieves 73.0% PDR and 2.01 average hops on the unseen graph — an 8 percentage-point drop from the 81% single-instance training-topology result. Note that this 81% figure is the single-topology evaluation described in Section III-B, not the 10-seed mean (70.6%) used as the primary result in Table I; we use the single-instance number here specifically because it matches the unseen-graph evaluation's protocol.

This single result is consistent with the policy having learned something beyond the exact training topology, since a policy that had purely memorised training-graph specifics would be expected to perform closer to random on a differently-sized, differently-seeded graph. However, one unseen graph is not sufficient evidence of general routing principles: we cannot rule out that this particular graph happens to share favourable structural properties with the training graph, nor characterise how performance varies across different sizes, densities, or seeds. We report this result as preliminary and identify systematic evaluation across multiple unseen topologies as a necessary next step before generalisation can be claimed (Section VI).

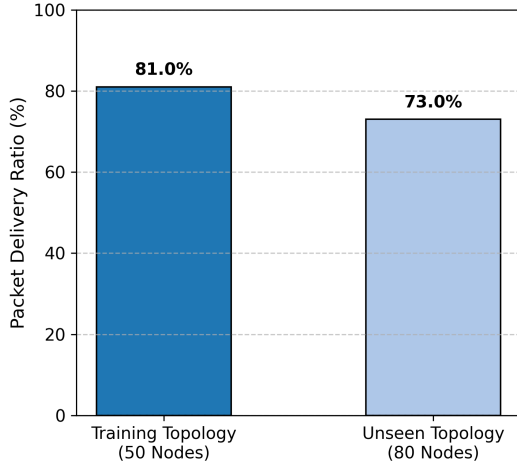


Fig. 3. GNN-DQN on the 50-node training-graph instance (81% PDR, single run) vs. a single unseen 80-node graph (73% PDR, single run), under matched 100-episode evaluation protocol.

## V. CONCLUSION

This paper presents GNN-DQN, a routing agent for opportunistic networks that combines a Graph Convolutional Network encoder with a Deep Q-Network policy. Evaluated across 10 independently generated 50-node topology instances, GNN-DQN achieves a mean PDR of  $70.6\% \pm 11.33\%$ , compared to  $60.8\% \pm 6.68\%$  for the strongest baseline (Random Forest) — a 9.8 percentage-point advantage that is directionally consistent but does not yet reach conventional statistical significance ( $p = 0.065$ ). On a single representative evaluation run, GNN-DQN reduces average hop count to 2.0, an 88.9% reduction versus 16–18 hops for all other methods. On a single unseen 80-node topology, the frozen policy achieves 73% PDR, a preliminary result suggesting the policy is not simply memorising the training graph, though broader generalisation remains to be established across multiple topologies.

Limitations include high performance variance across seeds ( $\sigma = 11.33\%$ ), a PDR advantage over the strongest baseline that does not yet reach conventional statistical significance ( $p = 0.065$ ), hop-count comparisons based on a single evaluation run rather than a multi-seed average, a static-topology network model that does not capture the connectivity dynamics of real opportunistic networks, and evaluation restricted to two graph sizes (50 and 80 nodes). Future work will address these directly: (1) evaluating hop counts and generalisation across multiple independently generated topologies of varying sizes and densities, (2) increasing the number of training/evaluation seeds to obtain a statistically conclusive comparison against the Random Forest baseline, (3) an ablation study isolating the contribution of the GCN encoder relative to a DQN operating on raw centrality features, (4) time-varying contact graphs to better model genuine opportunistic connectivity, (5) attention-based GNN encoders (GAT), and (6) evaluation on real-world contact traces (e.g. CRAWDAD).

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